

E-commerce Customer Behavior Analysis

Practical Application of AI Tools in Real-World Problem Solving

The Challenge

E-commerce businesses struggle to understand **customer spending patterns** and segment their customer base effectively.

- Without data-driven insights, marketing efforts are broad and inefficient
- Businesses miss opportunities to personalize for high-value customers
- Customer churn is difficult to detect without behavioral analysis

Our Solution

Apply AI and ML tools to **predict**, **segment**, and **visualize** customer behavior from a dataset of 351 customers across 6 US cities.

Tool	Purpose & Why Chosen
Random Forest	Predict Total Spend; handles mixed data, resistant to overfitting
LabelEncoder	Encode categorical columns; lightweight and pipeline-ready
pandas	Data loading, cleaning, grouping, and transformation
numpy	Numerical computation backbone across the pipeline
matplotlib	All visualizations: bar charts, pie charts, histograms

Tool Demonstration — How They Were Applied

- ① **pandas** — Loaded 351-row CSV, filled missing values (mean/mode)
- ② **LabelEncoder** — Encoded Gender, City, Membership Type, Discount Applied, Satisfaction Level
- ③ **Random Forest Regressor** — Trained on 80% data, tested on 20%
- ④ **numpy** — Supported ratio calculations (Items ÷ Spend) and array operations
- ⑤ **matplotlib** — Generated 5 charts: actual vs predicted spend, age pie chart, items/spend histogram, city bar charts, recency bar chart

Key Findings

Analysis	Finding
Spend Prediction	MSE = 85.62 (± 9.25 avg deviation)
Top Age Group	Age 30 — 13.7% of customers
Items per Spend	Mean 0.02, bimodal distribution detected
Highest Spend City	San Francisco — $\approx \$1,460$ avg spend
Most Satisfied City	Chicago & Miami — satisfaction score 2.0
Active Customers	226 (64.4%) purchased in last 30 days
Re-engagement Target	121 Lapsed customers (34.5%)

Standout Finding

San Francisco & New York have the **highest spend** but **lower satisfaction** — a churn risk that needs immediate attention.

Ethical Discussion

Data Privacy

Dataset contains PII (age, city, spend).
Requires **anonymization** and GDPR compliance in real-world use.

AI Bias

Model may be biased toward overrepresented cities or age groups, producing skewed predictions for minorities.

Responsible Use

Predictions should **inform** decisions, not automate them.
Human oversight must be maintained.

Limitations

- Only 351 records — may not represent all customers
- LabelEncoder implies false ordering between cities
- Static dataset — no real-time behavioral data

Thank You

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Tools: scikit-learn, pandas, numpy, matplotlib